Chapter 1 — The Foundations: Logic and Proofs

The whole chapter on one slide: what you must be able to state, build, and prove.

Rosen, 8th ed. · §1.1–1.8

1 · Propositions

A **proposition** is a declarative sentence that is either true or false — never both, never neither.

✓ $4+2=7$ (false, still a proposition) · "Cairo is in Egypt"
✗ "Close the door." · "$x+1=2$" · "This sentence is false."

2 · The five connectives

Symbol	Name	Read	False exactly when
$\neg p$	negation	"not p"	p is true
$p \wedge q$	conjunction	"and"	either is false
$p \vee q$	disjunction	"or" (inclusive)	both are false
$p \oplus q$	exclusive or	"either … or, not both"	p, q agree
$p \to q$	conditional	"if p then q"	**T → F only**
$p \leftrightarrow q$	biconditional	"p iff q"	p, q differ

Precedence: $\neg$, then $\wedge$, then $\vee$, then $\to$, then $\leftrightarrow$. A formula on n variables has 2^n rows.

3 · The conditional family

From $p \to q$:

Name	Form	Equivalent to original?
converse	$q \to p$	no
inverse	$\neg p \to \neg q$	no
contrapositive	$\neg q \to \neg p$	yes

"p only if q" means $p \to q$ — **not** $q \to p$.

4 · Classifying a formula

Build the truth table and read the last column: **tautology** — all **T** · **contradiction** — all **F** · **contingency** — a mix.
$p \equiv q$ means $p \leftrightarrow q$ is a tautology.

5 · The equivalences you must know

$\neg(p \wedge q) \equiv \neg p \vee \neg q$	De Morgan
$\neg(p \vee q) \equiv \neg p \wedge \neg q$	De Morgan
$p \to q \equiv \neg p \vee q$	conditional as $\vee$
$\neg(p \to q) \equiv p \wedge \neg q$	negating "if–then"
$p \to q \equiv \neg q \to \neg p$	contrapositive
$p \leftrightarrow q \equiv (p \to q) \wedge (q \to p)$	biconditional

Also: identity, domination, idempotent, double negation, commutative, associative, distributive, absorption.

6 · Predicates and quantifiers

$P(x)$ has no truth value until x is given a value — or bound by a quantifier.

Quantifier	True when	Disproved by
$\forall x P(x)$	$P(x)$ holds for every x	one counterexample
$\exists x P(x)$	$P(x)$ holds for some x	checking all x fail
$\exists! x P(x)$	exactly one such x	zero or two witnesses

De Morgan: $\neg\forall x P(x) \equiv \exists x \neg P(x)$, $\neg\exists x P(x) \equiv \forall x \neg P(x)$ — push $\neg$ inward, flipping each quantifier.

"All S are P" is $\forall x(S(x) \to P(x))$, **never** $\forall x(S(x) \wedge P(x))$.
In nested quantifiers the **order matters**: $\forall x \exists y \neq \exists y \forall x$.

7 · Rules of inference

An argument is **valid** when the premises cannot all be true while the conclusion is false.

Rule	Premises	Conclusion
modus ponens	$p \to q,\ p$	$\therefore q$
modus tollens	$p \to q,\ \neg q$	$\therefore \neg p$
hypothetical syll.	$p \to q,\ q \to r$	$\therefore p \to r$
disjunctive syll.	$p \vee q,\ \neg p$	$\therefore q$
resolution	$p \vee q,\ \neg p \vee r$	$\therefore q \vee r$
addition / simplification	p $\quad p \wedge q$	$\therefore p \vee q$ $\quad \therefore p$

Fallacies: affirming the conclusion ($p \to q, q \therefore p$) and denying the hypothesis ($p \to q, \neg p \therefore \neg q$).

8 · The three proof methods

To prove $p \to q$:

Method	Assume	Derive
direct	p	q
contraposition	$\neg q$	$\neg p$
contradiction	$p \wedge \neg q$	a falsehood

Choosing: start direct; switch to contraposition when $\neg q$ is the concrete hypothesis (e.g. "n^2 odd → n odd"); use contradiction for non-existence and irrationality ($\sqrt{2} \notin \mathbb{Q}$). n even $\equiv n = 2k$; n odd $\equiv n = 2k+1$.

Write these from memory

1. The truth table of $\to$ — false **only** in the row **T → F**.
2. De Morgan's laws, for connectives **and** for quantifiers.
3. Modus ponens, modus tollens, hypothetical syllogism.